Name : Fahad Ali

Roll no: 23P-0569

Section : CS-4k
Operating system Lab

Revision Lab Task

Task:1

codes snapshot:

Open

libraray.h
~/Desktop/Revisionlab

```c

typedef struct {
    int id;
    char name[100];
} Book;

// Function prototypes.......
void add_book(Book *book);
void issue_book(int book_id, const char *student_name);

```

Open

book.c
~/Desktop/Revisionlab

```c

#include <stdio.h>
#include <string.h>
#include "libraray.h"


void add_book(Book *book) {
    printf("Enter book ID: ");
    scanf("%d", &book->id);
    getchar();  // Consume newline character
    printf("Enter book name: ");
    fgets(book->name, sizeof(book->name), stdin);
    book->name[strcspn(book->name, "\n")] = '\0';  // Remove newline at the end
    printf("Book '%s' with ID %d has been added.\n", book->name, book->id);
}

```

Open

issue.c
~/Desktop/Revisionlab

```c
#include <stdio.h>
#include <string.h>
#include "libraray.h"


void issue_book(int book_id, const char *student_name) {
    printf("Book with id %d has issued to %s.\n", book_id, student_name);
}

```

Open

main.c
~/Desktop/Revisionlab

```c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include "libraray.h"

#define MAX_BOOKS 100

int main() {
    Book books[MAX_BOOKS];
    int choice, book_count = 0;
    int book_id;
    char student_name[100];

    while (1) {
        printf("\nLibrary Management System\n");
        printf("1.Add Book\n");
        printf("2.Issue Book\n");
        printf("3.Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);
        getchar();
        --
        switch (choice) {
            case 1:
                if (book_count < MAX_BOOKS) {
                    add_book(&books[book_count]);
                    book_count++;
                } else {
                    printf("Cannot add more books, library is full.\n");
                }
                break;
            case 2:
                printf("Enter book ID to issue: ");
                scanf("%d", &book_id);
                getchar();
                printf("Enter student name: ");

                fgets(student_name, sizeof(student_name), stdin);
                student_name[strcspn(student_name, "\n")] = '\0';
                issue_book(book_id, student_name);
                break;
            case 3:
                printf("Exiting system...\n");
                exit(0);
```

C

Make file snapshot:

Makefile
~/Desktop/Revisionlab

```

library: main.c book.c issue.c
	gcc main.c book.c issue.c -o library


run: library
	./library


clean:
	rm -f library

```

Output:

student@Lab5-Pc: ~/Desktop/Revisionla

```
student@Lab5-Pc:~/Desktop/Revisionlab$ make
gcc main.c book.c issue.c -o library
student@Lab5-Pc:~/Desktop/Revisionlab$ make run
./library

Library Management System
1.Add Book
2.Issue Book
3.Exit
Enter your choice: 1
Enter book ID: 345
Enter book name: os
Book 'os' with ID 345 has been added.

Library Management System
1.Add Book
2.Issue Book
3.Exit
Enter your choice: 2
Enter book ID to issue: 2
Enter student name: 345
Book with id 2 has issued to 345.

Library Management System
1.Add Book
2.Issue Book
3.Exit
Enter your choice: 3
Exiting system...
student@Lab5-Pc:~/Desktop/Revisionlab$
```

```
student@Lab5-Pc:~/Desktop/Revisionlab$ make clean
rm -f library
student@Lab5-Pc:~/Desktop/Revisionlab$
```

Task:2

```
student@Lab5-Pc:~$ cd Desktop
student@Lab5-Pc:~/Desktop$ mkdir Project_Audit
student@Lab5-Pc:~/Desktop$ cd Project_Audit
student@Lab5-Pc:~/Desktop/Project_Audit$ mkdir Reports Logs
student@Lab5-Pc:~/Desktop/Project_Audit$ ls
Logs  Reports
student@Lab5-Pc:~/Desktop/Project_Audit$
```

```
student@Lab5-Pc:~/Desktop/Project_Audit$ pwd
/home/student/Desktop/Project_Audit
student@Lab5-Pc:~/Desktop/Project_Audit$
```

```
student@Lab5-Pc:~/Desktop/Project_Audit$ cd Reports
student@Lab5-Pc:~/Desktop/Project_Audit/Reports$ touch report1.txt report2.txt summary.txt
student@Lab5-Pc:~/Desktop/Project_Audit/Reports$ ls
report1.txt  report2.txt  summary.txt
student@Lab5-Pc:~/Desktop/Project_Audit/Reports$
```

```
student@Lab5-Pc:~/Desktop/Project_Audit/Reports$ pwd
/home/student/Desktop/Project_Audit/Reports
student@Lab5-Pc:~/Desktop/Project_Audit/Reports$
```

```
student@Lab5-Pc:~/Desktop/Project_Audit/Reports$ date >> report1.txt
student@Lab5-Pc:~/Desktop/Project_Audit/Reports$ date >> report2.txt
student@Lab5-Pc:~/Desktop/Project_Audit/Reports$ date >> summary.txt
student@Lab5-Pc:~/Desktop/Project_Audit/Reports$ echo "OS System is initialized successfully..." >> report1.txt
student@Lab5-Pc:~/Desktop/Project_Audit/Reports$ echo "An error detected in this" >> report1.txt
student@Lab5-Pc:~/Desktop/Project_Audit/Reports$ echo "User is login successfuly....." >> report1.txt
student@Lab5-Pc:~/Desktop/Project_Audit/Reports$ echo "An error occurred during system  processing." >> report1.txt
student@Lab5-Pc:~/Desktop/Project_Audit/Reports$ echo "Now Backup completed without error....." >> report1.txt
student@Lab5-Pc:~/Desktop/Project_Audit/Reports$
```

```
student@Lab5-Pc:~/Desktop/Project_Audit/Reports$ echo "Audit process started." >> report2.txt
echo "No error found in financial records." >> report2.txt
echo "Audit process completed successfully." >> report2.txt
student@Lab5-Pc:~/Desktop/Project_Audit/Reports$
```

```
student@Lab5-Pc:~/Desktop/Project_Audit/Reports$ grep "error" report1.txt > summary.txt
student@Lab5-Pc:~/Desktop/Project_Audit/Reports$
```

```
student@Lab5-Pc:~/Desktop/Project_Audit/Reports$ wc report1.txt report2.txt
  8  42 295 report1.txt
  4  20 130 report2.txt
 12  62 425 total
student@Lab5-Pc:~/Desktop/Project_Audit/Reports$
```

```
 12  62 425 total
student@Lab5-Pc:~/Desktop/Project_Audit/Reports$ ls -l
total 12
-rw-rw-r-- 1 student student 295 Feb 13 09:23 report1.txt
-rw-rw-r-- 1 student student 130 Feb 13 09:23 report2.txt
-rw-rw-r-- 1 student student 191 Feb 13 09:25 summary.txt
student@Lab5-Pc:~/Desktop/Project_Audit/Reports$
```

```
student@Lab5-Pc:~/Desktop/Project_Audit/Reports$ chmod 744 summary.txt
student@Lab5-Pc:~/Desktop/Project_Audit/Reports$ ls -l summary.txt
-rwxr--r-- 1 student student 191 Feb 13 09:25 summary.txt
student@Lab5-Pc:~/Desktop/Project_Audit/Reports$
```

```
student@Lab5-Pc:~/Desktop/Project_Audit/Reports$ pwd
/home/student/Desktop/Project_Audit/Reports
student@Lab5-Pc:~/Desktop/Project_Audit/Reports$
```

```
student@Lab5-Pc:~/Desktop/Project_Audit/Reports$ date
Fri Feb 13 09:30:44 AM PKT 2026
student@Lab5-Pc:~/Desktop/Project_Audit/Reports$
```

```
student@Lab5-Pc:~/Desktop/Project_Audit/Reports$ df -h
Filesystem      Size  Used Avail Use% Mounted on
tmpfs           1.6G  2.0M  1.6G   1% /run
/dev/sda6       134G   12G  116G  10% /
tmpfs           7.7G   46M  7.7G   1% /dev/shm
tmpfs           5.0M  4.0K  5.0M   1% /run/lock
efivarfs        374K  280K   90K  76% /sys/firmware/efi/efivars
/dev/sda1        96M   50M   47M  52% /boot/efi
tmpfs           1.6G  140K  1.6G   1% /run/user/1000
student@Lab5-Pc:~/Desktop/Project_Audit/Reports$
```

```
student@Lab5-Pc:~/Desktop/Project_Audit/Reports$ ls
report1.txt  report2.txt  summary.txt
student@Lab5-Pc:~/Desktop/Project_Audit/Reports$
```

Task:3

```
student@Lab5-Pc:~/Desktop/Task3$ ls
result.sh
student@Lab5-Pc:~/Desktop/Task3$ chmod +x result.sh
student@Lab5-Pc:~/Desktop/Task3$ ./result.sh
Enter the name of student: fahad
Enter marks for subject 1: 60
Enter marks for subject 2: 70
Enter marks for subject 3: 75
Student Name is : fahad
Total Marks are : 205
Average is : 68
Grade   is : C
student@Lab5-Pc:~/Desktop/Task3$
```

Task:4

script snapshot:

```
#!/bin/bash

echo "Script Name is: $0"

if [ $# -eq 0 ]; then
    echo "No arguments were provided.plz provide arguments.."
else

    echo "Total Number of Arguments: $#"


    echo "All Arguments: $@"


    if [ ! -z "$1" ]; then
        echo "First Argument: $1"
    fi
    if [ ! -z "$2" ]; then
        echo "Second Argument: $2"
    fi
fi

echo "Current User: $USER"
echo "Home Directory: $HOME"
echo "Present Working Directory: $PWD"
```

Output:

```
student@Lab5-Pc: ~/Desktop/Task4
student@Lab5-Pc:~/Desktop/Task4$ chmod +x analyze.sh
student@Lab5-Pc:~/Desktop/Task4$ ./analyze.sh fahad shahzaib Hammad
Script Name is: ./analyze.sh
Total Number of Arguments: 3
All Arguments: fahad shahzaib Hammad
First Argument: fahad
Second Argument: shahzaib
Current User: student
Home Directory: /home/student
Present Working Directory: /home/student/Desktop/Task4
student@Lab5-Pc:~/Desktop/Task4$
```

Output with no arguments:

```
student@Lab5-Pc:~/Desktop/Task4$ ./analyze.sh
Script Name is: ./analyze.sh
No arguments were provided.plz provide arguments..
Current User: student
Home Directory: /home/student
Present Working Directory: /home/student/Desktop/Task4
student@Lab5-Pc:~/Desktop/Task4$
```